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ORIGINAL ARTICLE

A simple on-field fast knee-flexion test to assess acute knee flexor fatigue

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Abstract

Purpose In a practical setting, outside the laboratory, acute muscle fatigue may be underestimated because substantial recovery occurs during the elapsed time between the end of exercise and fatigue assessment. We introduce a simple field test to assess knee flexor contractile function quickly after exercise cessation.

Methods Fourteen young amateur football players performed maximally fast knee flexions (FKFs) in the prone position with their dominant leg, before (pre) and 20 s after finishing a series of fourteen fatiguing 40 m sprints (post) and again following 6 min recovery (rec). Peak angular acceleration (PAA) about the knee joint was measured with a small inertial measurement unit (IMU) firmly attached to the shin.

Results Although participants only practiced the FKFs for 1 min in the warm-up, the reliability of PPA was good with coefficients of variation of 3.0% (pre), 2.7% (post), and 3.6% (rec). Sprint time increased from 5.96 ± 0.40 s to 6.55 ± 0.37 s ($p < 0.001$, $f = 0.89$), PAA decreased from 107.1 ± 11.5 rad.s^{-2} to 94.1 ± 11.7 rad.s^{-2} ($p < 0.001$, $f = 0.50$) and following recovery ($p < 0.05$) values were 6.15 ± 0.39 s and 103.1 ± 10.7 rad.s^{-2}, respectively. The percentage decrease in PAA during FKFs was linearly related ($r^2 = 0.48$, $p = 0.01$) to the percentage increase in 40 m sprint time. In addition, PAA (pre) was related to the time of the first sprint ($r^2 = 0.33$, $p = 0.03$).

Conclusion The proposed FKF test is reliable and can easily be executed to evaluate acute knee flexor muscle fatigue on the field. The presented relations between (changes in) sprint performance and peak knee angular accelerations during isolated fast knee flexions are promising but need confirmation in larger-scaled studies.

Keywords Rapid force development · Football · Sprinting · Muscle fatigue · Butt-kicks

Abbreviations

ANOVA	Analysis of variance
CV	Coefficient of variation
d	Cohen's d effect size
f	Cohen's f effect size
EMG	Electromyography
FKF(s)	Fast knee flexion(s)
ICC	Intra class correlation coefficient
IMU	Inertial measurement unit
PAA	Peak angular acceleration
RPE	Rating of perceived exertion

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Introduction

Hamstring injuries present a great problem in sports, hampering athletes' careers and posing clubs with the financial costs of rehabilitation and missed matches, its prevalence in football has increased during recent seasons (Ekstrand et al. 2022). Hamstring injuries predominantly occur during sprinting and their incidence has been related to fatigue (Woods et al. 2004; Ekstrand et al. 2011; Della Villa et al. 2023). This has motivated researchers to study neuromuscular knee flexor fatigue outside the laboratory in a practical setting (Marshall et al. 2014; Wilmes et al. 2021). The study of fatigue is also interesting because of its relationship with performance decline. Fatigue has been

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defined as 'any reduction in the maximal capacity to generate force or power output' (Vollestad 1997), operationalized in the present study for the knee flexors as a reduction of 'unloaded' angular knee flexion acceleration and for sprint runs as an increase in sprint time. In football, knee flexor muscle contractile function deteriorates at half-time and post-matches during 11 versus 11 players and simulated matches (Wilmes et al. 2021; Marshall et al. 2014; Rahnama et al. 2003).

A major limitation of assessing neuromuscular fatigue in applied settings is the time it takes to place players in equipment used to measure contractile muscle function. Probably, acute fatigue is largely caused by metabolic changes in the muscle (Sundberg and Fitts 2019) and is known to show quick recovery (de Ruiter et al. 2000; Bishop et al. 2011; Jones 2010). With simulated matches and using a special on-field dynamometer, we recently measured knee flexor force 60 s after the last activity (Wilmes et al. 2021), other studies do not always explicitly report this time (Rahnama et al. 2003; Marshall et al. 2014), but 60 s will approximately be the lower limit. This may lead to underestimated fatigue, especially following intensive play, because recovery can be substantial in 60 s (de Ruiter et al. 2000; Bishop et al. 2011; Jones 2010). Thus, although decreased post-match knee flexor muscle strength has been reported (Rahnama et al. 2003; Marshall et al. 2014; Wilmes et al. 2021), it seems plausible that fatigue was underestimated in these studies. Since hamstring injuries are highly prevalent during running (Woods et al. 2004) and maximal sprinting (Askling et al. 2007), the risk of sustaining an acute hamstring injury might be greater during periods of intense play especially in the fatigued state (Woods et al. 2004; Ekstrand et al. 2011; Della Villa et al. 2023). Therefore, it would be valuable to assess fatigue, immediately following intense periods of play, when the metabolic milieu and muscle contractile capacity may differ substantially from those measured one to several minutes later (Bogdanis et al. 1995; Karatzaferi et al. 2001).

In the present study, we propose using on-field fast knee flexions (FKFs) executed in a prone position to assess knee flexor contractile function rapidly following intense exercise. The knee flexion test only requires players to wear a small inertial measurement unit (IMU) on the shank without further equipment, while a second IMU on a timing gate will be used to assess sprint time in the present study.

We reasoned that peak angular acceleration (PAA) during maximally fast 'unloaded' knee flexion, represents the knee extensors' capacity for rapid force development necessary to get the mass of the lower leg into motion. Others have studied maximally fast unloaded contractions during seated knee extensions in the laboratory. These were also executed without additional external resistance and ended in a cushion to decelerate the lower limb (Houston et al. 1988; Andersen et al. 2005). Andersen et al. explained the importance of rotational acceleration of the lower leg because it affects both knee moment and knee flexion/extension velocity (Andersen et al. 2005). In maximally fast unloaded knee extension or flexion, the total knee moment is the sum of the moment due to gravitational forces and the moment due to inertial forces (i.e., the moment directly related to rotational acceleration). In addition, the rotational peak velocity is reached much earlier (after about 90 ms) in unloaded knee extension than peak muscle force (after about 300 ms). Therefore, fast force development is important for maximal rotational acceleration and knee power (i.e., the product of knee moment and rotational velocity). Houston et al. found a positive correlation ($r = 0.68$) between peak angular acceleration (reached in about 10 ms) and the percentage of fast twitch fibre area in the vastus lateralis muscle (Houston et al. 1988). Together these results illustrate the importance of contraction onset, which is in line with studies on the very early phase (40-100 ms) of isometric force development (Folland et al. 2014) that in addition was shown to be related to vertical jump performance (de Ruiter et al. 2007, 2006).

Since fatigue is known to reduce both the force and speed of muscle contraction, PAA is expected to decrease with fatigue, and importantly, especially with short-term fatigue, PAA should largely recover after a few minutes of rest (Bogdanis et al. 1995; de Ruiter et al. 2000; Jones 2010; Bishop et al. 2011). An important requirement of any test assessing acute muscle fatigue is a low intra-session variability. If a test is easy to perform, this will reduce within-participant variability and increase test sensitivity.

Based on pilot experiments, we hypothesize that maximally fast knee flexions in a prone position are easy to execute and standardize. Consequently, PAA expects to have a low coefficient of variation and changes in PAA can be used to assess sprint-induced knee flexor fatigue quickly following exercise. The maximal shortening velocity of isolated muscle contractions is known to decrease with fatigue (e.g. (de Ruiter et al. 2000)) and during fatiguing repetitive sprints speed, step frequency, and knee angular velocity decrease (e.g. (de Ruiter et al. 2022)). Therefore, we also anticipated PAA in the knee flexion test to decline with fatigue. In addition, the participants will differ in sprint capacity and fatigue resistance, therefore, we also hypothesize a significant relation between the increase in sprint time and the decrease in PAA among participants.

Methods

Participants

Initially, seventeen amateur football players from different teams volunteered for the study (15 males, and 2 females). However, due to technical problems, the IMU data of two

participants were incomplete and one participant sustained an ankle sprain during the measurements. Therefore, the final data set (one female) comprises 14 participants (age: 23.1 ± 2.3 years, body mass: 77.2 ± 6.0 kg, height: 1.85 ± 0.04 m) with an average playing experience of 14.0 ± 4.5 years. They participated after providing informed consent. Among the players were seven defenders, four midfielders, and three attackers. They were injury-free for at least three months before the measurements and did not sustain a hamstring injury in the last twelve months. They engaged in one to two training sessions and one match per week. The study was done following the Declaration of Helsinki (2013) and received approval from the local scientific and ethics review board (VCWE-2019-070R1). All participants were individually (separately) tested. The study was conducted on outdoor artificial turf and the surface was dry. Weather conditions varied with a mean temperature of 14 °C (range 8–22), a mean wind speed of 17.6 $km.h^{-1}$ (range 12–30), and humidity of 69.1% (51–89).

Preparation and warm-up

Procedures were once more explained on the test day. The preferred back leg of the participant when attaining a split stance position before a sprint start was defined as the dominant leg. The skin over the tibial bone of this leg was shaved for proper fixation of the shank-IMU after the warm-up and the participant put on a compression calf sleeve (Hansaplast Sport, Beiersdorf, Hamburg Germany) used to fixate the IMU. After this, a supervised warm-up (15–20 min) was conducted, which included 4 min of jogging with football-specific exercises, some basic stretching, and 4 min of strength exercises, during which the participants also practiced correct execution of maximally fast unloaded knee flexions ('butt-kicks'). The warm-up ended with three 45 m sprints passing through the mechanical timing gates (see below) executed with increasing (60%, 80%, and 100%) voluntary effort.

Inertial measurement units (IMUs)

Following the warm-up IMUs (MPU-9150, Invensense, San Jose, CA, USA) were switched on and time-synchronized with a mechanical thud. The IMUs continuously sampled 3D linear acceleration, 3D angular velocity, and 3D magnetic field strength at 500 Hz. The data was stored on an internal SD card. One IMU was fixated on the tibial bone of the dominant leg below the knee joint at about one-third of the distance between the knee and ankle while avoiding contact with muscles. First, a stretch of Leukotape (12 × 5 cm) was applied as a ground layer to which the IMU was firmly attached using a double-sided Nanotape, covered with Fixomull Stretch (12 × 5 cm) and covered by the compression calf sleeve. This was followed by a standard IMU-to-shank calibration procedure requiring participants to stand still and upright for 10 s to determine the longitudinal axis of the shank, followed by a thigh rise in the sagittal plane to determine the frontal axis as detailed by Wilmes et al. (Wilmes et al. 2020). The second IMU was fixated with double-sided tape to a mechanical timing gate 40 m from the start, detecting the finish moments from the peak in the IMU signals caused when the participants passed through its light 1 m long swivel as detailed elsewhere (de Ruiter and van Dieen 2019).

Sprint and fast knee flexion protocol

Five minutes following warm-up, participants executed 14 maximal 45 m-sprints starting every 50 s (5 s countdown started at t = 45 s), with the foot of the non-dominant leg at the starting line. Following deceleration after a cone placed at 45 m (time gate at 40 m), they jogged back to the start line. Continuous verbal encouragement urged participants to go all out. Six min and 20 s after the finish of sprint number 14 one additional sprint was executed to assess sprint recovery.

The first sequence of five FKFs (details below) was done 2 min before the start of the first sprint (pre), and a second sequence was started 20 s after the finish of sprint 14 (post). A third sequence started 5 min after the finish of sprint 14 to assess knee flexion recovery (rec). After that, the participant walked back to the start line and 6 min and 20 s after the 14th sprint, executed a final 15th sprint to assess sprint recovery. Before each FKF sequence, the Rating of Perceived Exertion (RPE) was recorded on a ten-point scale (Arney et al. 2019).

Details of fast knee flexion execution

Participants laid down 15 m after the timing gate in a prone position with straight legs, their feet in plantar flexion and their chin resting on their folded arms. From this relaxed position, they were instructed to flex the knee of the dominant leg as fast as possible: 'kick your butt as fast as possible' (Fig. 1). They were urged not to decelerate the lower leg: 'Let your heel bounce back from your buttocks'. All FKFs were video recorded in the sagittal plane with a smartphone (Samsung Galaxy S21, 30 Hz) placed at 1.5 m and 0.3 m in height. This was done to corroborate correct execution: a fluent fast knee flexion during which the upper leg had to remain in contact with the field and the participant's heel had to bounce back from the buttocks.

Fig. 1 Fast Knee Flexion execution. From left to right: FKFs start position, position after ~ 1 rad knee flexion and at maximal flexion hitting the buttocks

Data analysis

Custom-written Matlab (version R2022b, The Mathworks, Inc., Natick, MA, USA) scripts were utilized for data analysis. The orientation of the shank IMU was continuously estimated using a gradient descent Madgwick algorithm (Madgwick et al. 2011). Together with the data of the IMU-to-shank calibration procedure, the 3D orientation of the shank was derived. The complete procedure has been described elsewhere in detail (Wilmes et al. 2020). Raw IMU data and estimated angular shank speed about the frontal axis (sagittal body plane) were stored in Matlab struct configurations and further analysed with custom-written Python (Version 3.11) scripts. These Python scripts comprised the automatized detection and extraction of FKFs and sprints.

Shank rotational speed about the frontal knee axis was 12 Hz low pass filtered using a second-order Butterworth filter. Sprint start time was defined as the first moment the shanks' rotational speed became > 1 rad.s^{-1} during the start (the back leg always moves first when starting from a three-point position). Finish time was defined as the instance where the vector norm of the linear acceleration signals of the time-gate-IMU became > 20 m.s^{-2} (de Ruiter and van Dieen 2019).

Since during FKFs, the thigh remains horizontal, angular shank IMU speed is equivalent to angular knee flexion speed. The onset of the FKF movement was defined as the last moment before knee flexion where angular shank speed was < 0.05 rad.s^{-1}. The resting shank angle was defined as the mean shank angle during the 400 to 100 ms time interval before onset. Attempts were deemed valid if the deviation from the resting shank angle during the 40 ms time interval before onset was less < pi /180 radians (= 1 degree), confirming the absence of meaningful (counter-) movement before onset. Angular knee flexion acceleration was the time derivative of angular speed.

Statistical analysis

Because the FKF test is new, a priori power calculation ($\alpha = 0.05$, $\beta = 0.95$) was based on the expected sprint time effect using GPower 3.1 (Faul et al. 2007). We made conservative choices by including only the most interesting sprints (numbers 1, 13, 14, and 15) in this calculation and using a moderate effect size estimation ($f = 0.5$) compared to other studies (> 1.0) (van den Tillaar 2018; Jimenez-Reyes et al. 2019). The estimated number of participants was 10 but given the uncertainties with the repeatability of the FKF test and possible dropout, we included 17 participants. As indicated in the above we report the data of 14 participants. However, for the recovery sprint (number 15), for reasons unknown participant number 7 did not sprint maximally (Fig. 2) this data point was an outlier and excluded from the analysis.

The mean values of the three attempts (out of five) with the greatest PAA were used to quantify changes over time (pre, post, and after recovery).

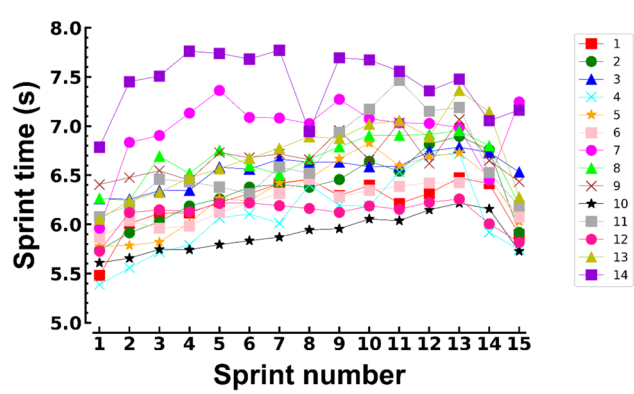

Fig. 2 Sprint times as a function of sprint number for individual participants (1–14). Sprint number 14 was the last sprint of the fatiguing series and number 15 was executed following 6 min recovery. Note that the time of sprint 8 of participant 14 isn't a measurement error, and participant number 7 did not sprint maximally following the 6-min recovery period

Normal distribution of the data was checked using Q-Q plots, histograms, and the Shapiro–Wilk test. To assess the statistical significance of fatigue and recovery, sprint numbers 1 (pre), 14 (post), and 15 (recovery) were included in separate repeated measures ANOVA for sprint times and PAA. For sprint time the pen-ultimate sprint (13) results were added in this analysis to investigate the presence of an end-spurt (Foster et al. 2023). On significance ($p < 0.05$) this was followed by Bonferroni corrected post hoc testing. Effect sizes are reported as Cohen's f and d values.

RPE was evaluated with the non-parametric Friedman test (Kendall's W is reported as rank correlation) followed by a Bonferroni-adjusted Wilcoxon test.

To obtain the reliability of the FKF test, for each of the three conditions separately (pre- and post-sprinting and following recovery), a two-way random Intra Class Correlation coefficient (ICC(2,1), agreement) and a coefficient of variation (CV) were calculated, using the three attempts with the greatest PAAs in each condition. CVs have been classified as highly reliable (≤5.0%), moderately reliable (5.1–10.0%), or unacceptable (> 10.0%). ICC values have qualitatively been described as poor (< 0.50), moderate (0.50–0.74), good (0.75–0.90), or excellent (> 0.90) (Koo and Li 2016).

Pearson's correlation coefficient was calculated to establish the significance of correlation and reported as coefficients of determination (r^2).

Results

RPE

The Rating of Perceived Exertion (RPE) increased ($W = 0.96$, $p < 0.001$) from a median value of 3 (range 2–5) before the sprint series to 9 (range 8–10) immediately after the fourteenth sprint ($p = 0.003$). RPE was not completely recovered following 6 min of rest (median 4, range 2–6, $p = 0.01$).

Sprint times

Most participants showed the expected gradual increase in sprint time from sprint 1 to 14. However, this pattern varied among participants (Fig. 2). ANOVA repeated measures (without participant 7 who did not sprint all-out after recovery) for the sprints of interest (1, 13, 14, and 15), showed a significant main effect of sprint number on 40 m times ($F_{3,36} = 56.7$, $p < 0.001$, $f = 0.89$). Mean (± standard deviations, n = 13) sprint times were 5.96 ± 0.40 s, 6.81 ± 0.40 s, 6.55 ± 0.37 s and 6.15 ± 0.39 s, respectively. All pairwise post hoc comparisons were significant, with effect sizes ranging from 1.06 to 2.85. Consistent with the end-spurt phenomenon, sprint 14 was faster than sprint number 13 ($p = 0.002$, d = 1.06). There was recovery as sprint 15 was faster than sprint 14 ($p = 0.001$, d = 1.43), but recovery was incomplete since sprint 15 still was slower than sprint 1 ($p = 0.002$, d = 1.41).

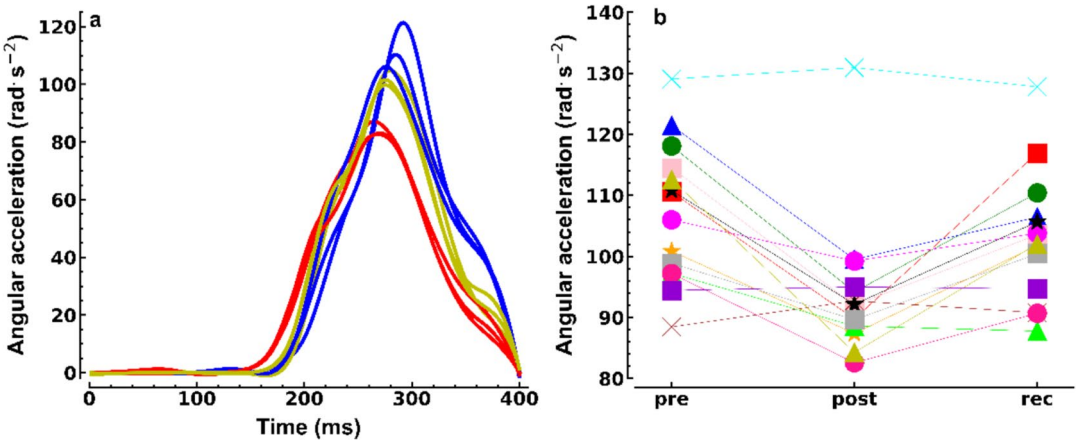

Fig. 3 **a** Example of knee angular acceleration as a function of time (participant 13). The three fastest Knee Flexion attempts pre- (blue), post-sprinting (red), and following recovery (olive) are depicted. Traces are aligned at $t = 400$ ms (zero crossings of the acceleration signals). **b** Individual ($n = 14$, for labels, see Fig. 2) peak knee angular accelerations pre-, post-sprinting, and following recovery (rec). Each data point is the mean value of the three attempts. Standard deviations are omitted for clarity

Peak angular acceleration during fast knee flexions

Figure 3a shows example data of the angular acceleration signals during the FKFs. ICCs (95% confidence interval) pre-sprints, post-sprints, and following recovery were 0.82 (0.44–0.93), 0.86 (0.48–0.95), and 0.79 (0.48–0.91), respectively. The CVs in the three conditions were 3.0%, 2.7%, and 3.6%.

PAA (rad.s^{-2}) significantly decreased ($F_{2,26}=16.2$, $p<0.001$, $f=0.50$) from 107.1 ± 11.5 (pre) to 94.1 ± 11.7 post-sprinting ($p=0.001$, d = 1.26) and it recovered to 103.1 ± 10.7 ($p=0.006$, d = 1.0). Although recovery values were not significantly different from the pre-sprint values ($p=0.08$, d = 0.66), recovery appeared incomplete, but less so than for the sprint times. Similar to changes in sprint time, the sprint-induced changes in PAA during the FKFs also varied considerably among participants (Fig. 3b).

The knee angle at which PAA (rad) was obtained significantly increased ($F_{2,26}=8.0$, $p=0.002$, $f=0.15$) from 0.31 ± 0.09 (pre) to 0.34 ± 0.10 post-sprinting ($p=0.02$, d = 0.86) and 0.34 ± 0.09 ($p=0.04$, d = 0.77) following recovery.

As hypothesized, there was a significant relation between the fatigue-induced decreases in sprint performance (increased sprint times between sprint 1 and 14) and the decreases in PAA during the FKFs (Fig. 4a, $p=0.01$, $r^2=0.48$, excluding the female player: $p=0.02$, $r^2=0.41$).

There also was a significant linear relation (Fig. 4b, $p=0.03$, $r^2=0.33$, excluding the female player: $p=0.08$, $r^2=0.26$) between sprint and FKF performance in the unfatigued state.

Discussion

The current findings are in line with our hypothesis. The results corroborate FKFs in a prone position being easy to perform and to standardize. Moreover, the decrease in sprint times during repetitive sprints varied among participants and was related to the decline in peak knee angular knee acceleration (Fig. 4a, $r^2=0.48$).

The low CVs (~ 3%) of PAA, despite participants being unfamiliar with the FKF test and only practicing during the warm-up, is a very promising finding. Moreover, only 3 out of 210 (= 14 × 5x3) attempts were invalidated and this was for lifting the upper leg from the ground. Good within-session reproducibility of any test aiming to quantify changes during a training session or match is important because this enables the detection of relatively small changes in the variable of interest. The FKFs proved sensitive enough to detect a decrease of 12.0 ± 9.1% in PPA.

The test seems an asset in quantifying on-field knee flexor muscle fatigue. Similarly, vertical jumps have been proposed to assess on-field muscle fatigue rapidly after exercise (Jimenez-Reyes et al. 2019). However, jumps involve the coordinated contractions of several muscle groups, with knee extensors, gluteal, hamstring, and calf muscles all contributing to jump performance (Bobbert and van Ingen Schenau 1988). The FKF test specifically targets the knee

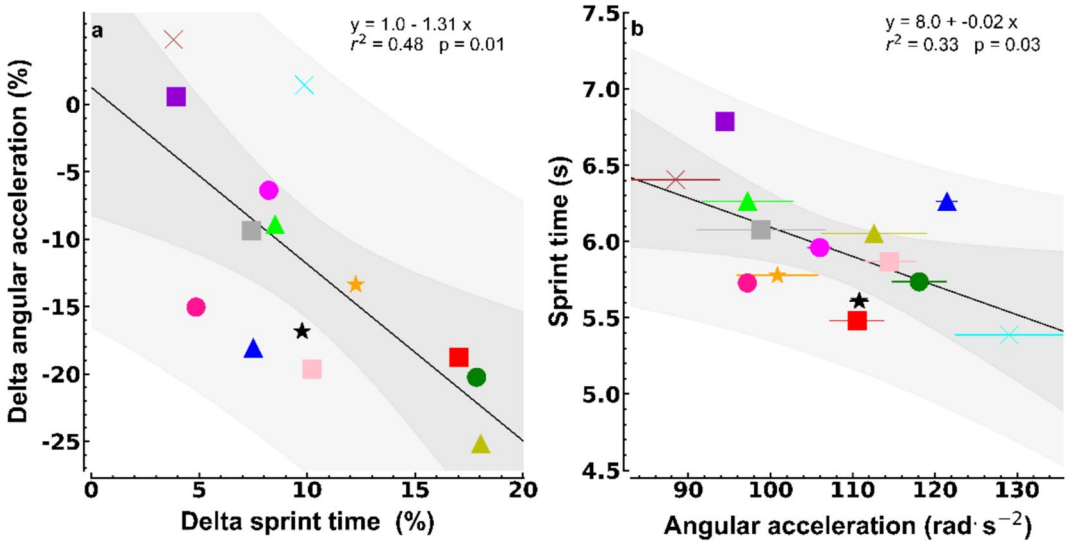

Fig. 4 **a** Pre- to post-sprint changes in peak knee angular acceleration during FKFs as a function of the percentage changes in 40 m sprint time (x-axis). The Pre-sprint values are set to 100%. **b** Time of the first sprint as a function of pre-sprint peak knee angular acceleration during FKFs (mean values of three fastest attempts ± standard deviations) on the x-axis. Regression lines are shown with 95% confidence (dark grey) and prediction (light grey) intervals

flexors. We tried to limit calf muscle contribution to the knee flexions by starting with the foot in plantar flexion, but clearly, other muscles are involved in stabilization. Deformations of the vastus lateralis muscle were visible in the video recordings at the start of the knee flexions. The gluteal muscles may also be involved in movement stabilization. The extent of these co-contractions needs further investigation, but we are confident that the hamstring muscle group is the main determinant for FKF performance.

Hamstring injuries have been linked to fatigue and occur mostly acutely during running (Della Villa et al. 2024). Therefore, monitoring the state of knee flexor fatigue and subsequently adapting the training load or extending the recovery periods within and between sessions might reduce injury risk. It is impossible to assess the metabolic or contractile status of the knee flexors *during* sprinting, and following exercise recovery usually is rapid (Bogdanis et al. 1995; de Ruiter et al. 2000; Jones 2010; Bishop et al. 2011). Note that although the angle at which PAA was reached did not return ($p < 0.05$) to its pre-sprint value, a change of about 1.7 degrees probably is not meaningful.

The strength of the presented FKF test is, that in contrast to dynamometer testing, the fatigue assessment can be done within a handful of seconds after vigorous exercise on the field. In the present study, for reasons of standardization, we used a conservative 20 s after crossing the finish line of the last sprint. We assumed that recovery at this moment was still limited. Angular acceleration largely recovered after 5-6 min of rest as did sprint time. With the present set-up, we can only speculate that the main reasons for the reductions in PAA and sprint performance probably were of peripheral origin and were caused by intramuscular metabolic changes like a decrease in phosphocreatine, and an increase in inorganic phosphate, an increased proton concentration (Sundberg and Fitts 2019), and potentially interstitial potassium concentration (Lindinger and Cairns 2021). In addition to the direct interference of these metabolic by-products with cross-bridge function and sarcoplasmic calcium ion release (Sundberg and Fitts 2019), these changes feedback on the central nervous system via group III and IV afferents, making it plausible that decreased neural activation also contributed to the reduction of angular acceleration (Broxterman et al. 2018). Full recovery after a vigorous football match may take (more than) 72 h (Krustrup et al. 2011; Carmona et al. 2024). Whether FKF performance is sensitive to longer-lasting aspects of fatigue remains to be established. This would necessitate measuring across days and good reliability of the FKFs between different days should, therefore, be established.

The percentage decrease in peak knee angular acceleration during FKFs after the fatiguing sprints accounted for 48% of the variation in the increase in sprint time. We think this is quite a high percentage because the sprinting technique varied considerably among our participants and their pacing strategies probably also affected the changes in sprint time. In well-trained and very motivated sprinters sprint time would increase about linearly with sprint number and the final sprint would be the slowest (Jimenez-Reyes et al. 2019). However, our participants represent the average Dutch amateur football player, and notably different patterns were seen as illustrated by participants 7 and 14 in Fig. 2. In all but participant number 6, the last sprint was faster than the penultimate sprint of the series. This is often seen (van den Tillaar 2018) and relates to the end-spurt phenomenon (Foster et al. 2023). In the present study, the 'end-spurt' varied considerably with the last sprint being about 0.06 s faster in participants 1,3, and 10 while others (4,7,9,11,14) were more than 0.41 s faster than their penultimate sprint. Therefore, although all participants indicated that they felt fatigued post-sprinting (RPE values ≥ 8), it is likely that the performance decreases were also affected by motivational factors. Speculatively, FKF performance is less affected by motivation and pacing since short-lasting isolated kicks are experienced as far less demanding compared to a maximal 40 m sprint. Taking the above differences in sprint execution into account, the relation between the increases in sprint time and the decrease in PAA during the FKFs (Fig. 4a) seems quite convincing. It remains to be established whether the present knee flexion test can quantify different levels of knee flexor fatigue across days within individual players.

We did not anticipate finding a significant relation between sprinting, involving coordinated muscle contractions of many muscle groups, and PAA of the unfatigued knee flexors in isolation ($r^2 = 0.33$). However, running speed is undoubtedly related to fast (lower) leg movements as good sprinters accommodate step frequencies of about 4.5 Hz (Debaere et al. 2013) necessitating very high knee angular speeds (Kivi et al. 2002). Thus, the presented relation between isolated maximal knee flexion acceleration and sprint speed may not be coincidental after all, however, it needs confirmation in larger studies preferably with sprint-trained athletes of different performance levels. In future studies, measuring EMG is recommended, since the capacity for maximal fast neuromuscular activation is an important factor in fast force development, and high PAAs expectantly are preceded by a high neural drive that would lead to high bursts of EMG activity at the onset of these explosive muscle contractions (de Ruiter et al. 2006; Tillin et al. 2010; Del Vecchio et al. 2019).

In conclusion, the proposed fast knee flexion test is reliable and can easily be executed to evaluate acute knee flexor muscle fatigue on the field. The presented relations between (changes in) sprint performance and peak knee

angular accelerations during isolated fast knee flexions are interesting but need confirmation in other studies.

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Data availability In addition to the supplementary materials, the data that support the findings of this study are available from the corresponding author (C.J. de R.) upon reasonable request.

Declarations

Conflict of interest The authors have no financial or non-financial interest to disclose.

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